Cubic-extension transfer: frozen evidence import

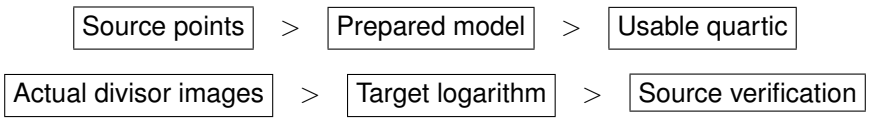
2026-10-25 Study: cubic_extension_transfer_20261025

Status. Historical SageMath 10.6 source and recorded certificates; no new research experiment, native performance result, or leaderboard promotion.

Question. When can a prime-order elliptic-curve logarithm over a cubic extension be transferred into a usable plane-quartic Jacobian over the base field, and what obstructs the twelve-candidate construction?

Recorded finite result. The retained example over the cubic extension of the field with 157 elements has source prime order 3868159. All twelve transverse targets have independently certified order 3871629. Two nonprepared sources each have a unique rational degree-three isogeny into that all-obstructed prepared model. A second step gives a usable target, and the retained complete solver recovers scalar 271828 on the original curve. This is a counterexample to a total-one-step guarantee within the specified construction. One extra step after torsion preparation remains open.

Structure. The recorded 109-field comparison has different coverage at the same geometric endomorphism discriminant, -21195. The complete finite CM screen contains twelve Frobenius orbits and one all-obstructed orbit. A degree-five isogeny repairs it while preserving the three-torsion type and endomorphism order. These statements have the source report's ordinary, non-subfield, prime-order and transverse-kernel hypotheses.



Recorded checks. Ten complete prototype transfer cases and seven targeted exceptional-input checks are retained. The saved verifier checks source scalars, exact line intersections, large-prime graph substitutions and modular systems. It is not a cross-host or independent-agent native audit.

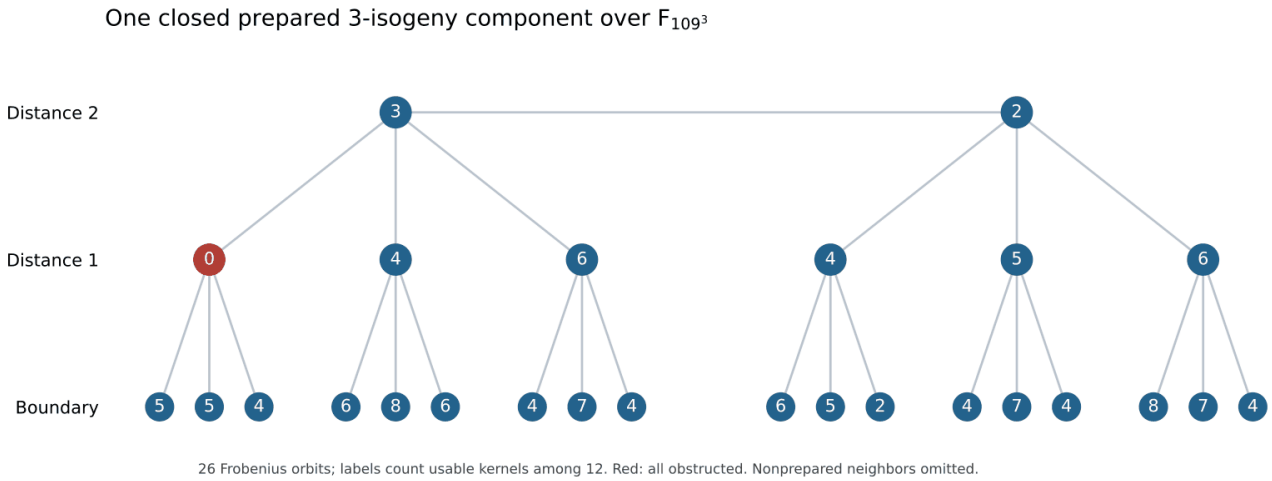
Timing limits. The frozen records used Sage and a simple source rho control; some jobs ran concurrently. They do not supply isolated native operation counts, the strongest folded rho, an ecbench audit, or external-tool qualification. The historical seconds are diagnostics; no cost ranking, new exponent, fastest-tool status or deployed-curve conclusion is admitted.

Publication. The source bundle contains 164 original files. This PR includes 159 byte-for-byte files and hash manifests for five original files awaiting unchanged S3 upload and readback. No remote availability is claimed. Conductor, Cargo/Rust and Lean/Lake are unavailable locally. Repository-wide release tests, native replay, faithful Lean bridges and full artifact hydration remain open. See PUBLICATION.json and the admission problem.

Source links. Relative to the study directory:

- historical/extension-field-experiments/research_lead/FOUR_TRACK_RESULTS.md
- results/forced_two_step_audit.json
  results/explicit_forced_two_step_witnesses.json under that research lead.
- results/forced_source_pipeline.json, results/endomorphism_audit.json, and results/cm_coverage.json.

Frozen prepared-component figure



This is the original study's component figure, rendered from its frozen SVG. It records 26 prepared Frobenius orbits over the cubic extension of the 109-element field. Labels count usable kernels among twelve; red marks zero coverage. Nonprepared neighbors are omitted. The exact models and edges are in results/neighborhood_component.json; the source and proof scope are in FOUR_TRACK_METHOD.md. This is a historical figure, not a new repository-standard volcano-renderer run or a geometric formalization.

The canonical scoreboard, registry, benchmark index and IC leaderboard are unchanged because the historical diagnostics have not met their admission gates. A later native study must freeze its inputs, isolate and fully price both arms, use the strongest applicable reference, independently replay the answers and satisfy the validation schema before a promotion.

Original bundle SHA-256:
6c6e079c8ebb64b52f03039bad743dc0948d0f1953b9f528b76e684959405aba